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Exploring Historical Information for RGBE Visual Tracking with Mamba

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Abstract

Combining the advantages of conventional and event cameras for robust visual tracing has drawn extensive interest. However, existing tracking approaches heavily engage in complex cross-modal fusion modules, leading to higher computational complexity and training challenges. Besides, these methods generally ignore the effective integration of historical information, which is crucial to grasping the change in the target's appearance and motion trends. Given the recent advancements in Mamba's long-range modeling and linear complexity, we explore its potential in addressing the above issues in RGBE tracking tasks. Specifically, we first propose an efficient fusion module based on Mamba, which utilizes a simple gate-based interaction scheme to achieve effective modality-selective fusion. This module can be seamlessly integrated into the encoding layer of prevalent Transformer-based backbones. Moreover, we further present a novel historical decoder that leverages Mamba's advanced long sequence modeling to effectively capture the target appearance changes with autoregressive queries. Extensive experiments show that our proposed approach achieves state-of-the-art performance on multiple challenging short-term and long-term RGBE benchmarks. Besides, the effectiveness of each key Mamba-based component of our approach is evidenced by our thorough ablation study.Code will be released at: https://github.com/scy0712/MamTrack

1. Introduction

Visual object tracking (VOT) is a fundamental task in computer vision[21–23, 30, 46] that focuses on estimating the position and shape of a target within video sequences based on its initial state [1, 3, 5, 17, 35, 36, 38, 47].Owing to the advantages of high temporal resolution and dynamic range offered by event cameras [33, 34, 40], there has been a growing trend of integrating them with conventional RGB cameras for robust visual tracking tasks [10, 28, 32, 41–43, 48]. These RGBE multimodal approaches can significantly enhance tracking performance in challenging conditions, such as fast motion and limited illumination. Although being effective, the above multimodal methods expend a substantial amount of labor on developing overly intricate fusion algorithms. For example, FENet [41] employs a total of 8 self-attention and 4 complex cross-domain attention operations to integrate event and frame features at various levels. Implementing these sophisticated strategies will significantly increase the computational complexity and challenge of the training.

Moreover, the complex fusion designs make existing RGBE methods usually overlook the mining and application of historical temporal information, which is crucial for tracking as it captures changes in the target's state. Several RGB-only trackers [1, 4, 38] have been developed to update appearance information by replacing or fusing the templates. Occasionally, these straightforward methods ineffectively filter out unreliable updated templates and thus compromise the performance. To address this issue, some

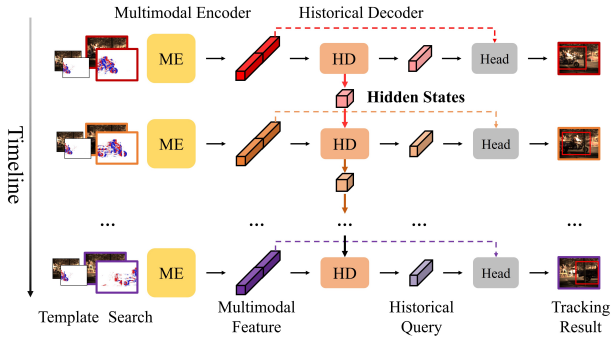


Figure 1. Overview of MamTrack and its two main components: (i) Multimodal Encoder is responsible for extracting and fusing features from RGB and event modalities; (ii) Historical Decoder is utilized to mine the temporal information of target appearance changes.

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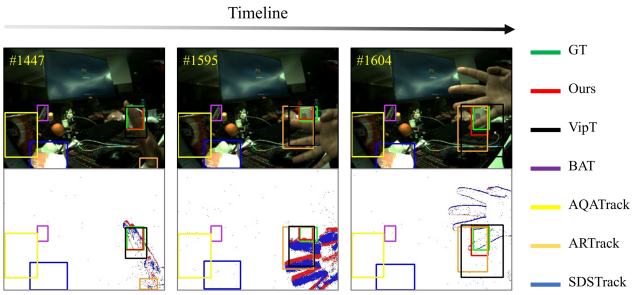


Figure 2. A comparison of our method with other recent SOTA trackers in long-term sequence. Benefiting from the proposed Historical Decoder, our method can effectively capture target appearance changes to guide robust long-term tracking even in partial occlusion scenes.

methods [35, 36] incorporate an autoregressive scheme into the Transformer to transfer temporal information across multiple search frames. However, the Transformer structure lacks intrinsic designs for temporal sequential modeling, resulting in an inherent constraint, particularly in handling long sequences.

Recently, Mamba [11], a novel State Space Model (SSM) [9, 12, 26, 37], has emerged as a highly promising approach for efficient long sequence modeling due to convolutional computation and linear complexity. Drawing inspiration from this advancement, as shown in Figure 1, we present mamba-based modules to facilitate robust RGB-E tracking by efficient multimodal fusion and effective historical information transmission. Specifically, we first propose an efficient fusion module based on Mamba, which utilizes a simple gate-based interaction scheme to enhance the relationship between search and template tokens, as well as the complementarity between RGB and event modalities. This module is then easily inserted into the encoding layer of prevalent transformer-based backbones for combining the advantages of self-attention. Furthermore, we present a novel historical information decoder, which investigates the great potential of mamba in long sequence modeling to capture the appearance of targets by introducing a set of additional autoregressive queries.

Extensive experiments on multiple challenging RGBE benchmarks show the effectiveness of our approach, which outperforms existing state-of-the-art (SOTA) methods in both short-term and long-term tracking (see Figure 2). Ablation studies verify the importance of each key component. More importantly, we reveal the considerable potential of Mamba architecture in object tracking tasks. In summary, we make the following contributions:

- We present a novel and effective framework, dubbed *MamTrack*, a pioneering approach to incorporate the Mamba mechanism into RGBE multimodal tracking tasks.
- Our proposed Mamba-based fusion module and historical decoder are capable of adaptively fusing multimodal information and effectively capturing the historical temporal cues of the target appearance changes, respectively.
- The proposed trackers achieve SOTA performance on multiple challenging short-term and long-term RGBE tracking benchmarks.

2. Related Works

2.1. Mamba

The concept of the State Space Model (SSM) was first introduced in the S4 model [12], presenting a distinctive architecture capable of effectively modeling global information compared to traditional convolutional neural networks and Transformers. In their general form, SSM can be written as follows:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t), \tag{1}$$
$$y(t) = C(t)x(t) + D(t)u(t). \tag{2}$$

This process maps a 1-D input sequence $u(t) \in \mathbb{R}$ to a 1-D output sequence $y(t) \in \mathbb{R}$ through an implicit N-D latent state sequence $x(t) \in \mathbb{R}^N$. Building on the S4 model, the S5 model [26] reduces complexity to a linear level, while H3 [9] introduces it into language model tasks. Mamba [11] introduces a data-dependent SSM layer and a selection mechanism using parallel scan (S6), achieving higher inference speed and overall metrics compared to equivalent-scale Transformers. Recently, Mamba has been employed in discriminative computer vision models and drawn great attention [7, 14, 18, 20, 49]. For example, ViM [49] applies Mamba to the Vision Transformer [8] architecture. VMamba [20] designs a cross-scan mechanism to bridge the gap between 1-D array scanning and 2-D plain traversing. Unlike previous efforts, we explore a novel architecture for RGB-E tracking based on Mamba, further unleashing its potential in visual tasks.

2.2. RGB-Event Single Object Tracking

Recently, due to the unreliable nature of RGB-only data in challenging scenarios, more and more studies introduced event cameras to enhance tracking performance. These multimodal trackers can be primarily categorised into two categories based on the vision backbone employed. The first one [29, 41, 42] utilizes conventional CNNs to extract features from both RGB and event modalities, followed by a meticulously designed cross-modal attention mechanism for fusion. The second approach [28, 32] adopts the Transformer as the backbone, leveraging the multi-head attention mechanism to effectively model the indispensable correlation relationship between the RGB and the event modalities. To tackle the challenge of limited multimodal tracking data,

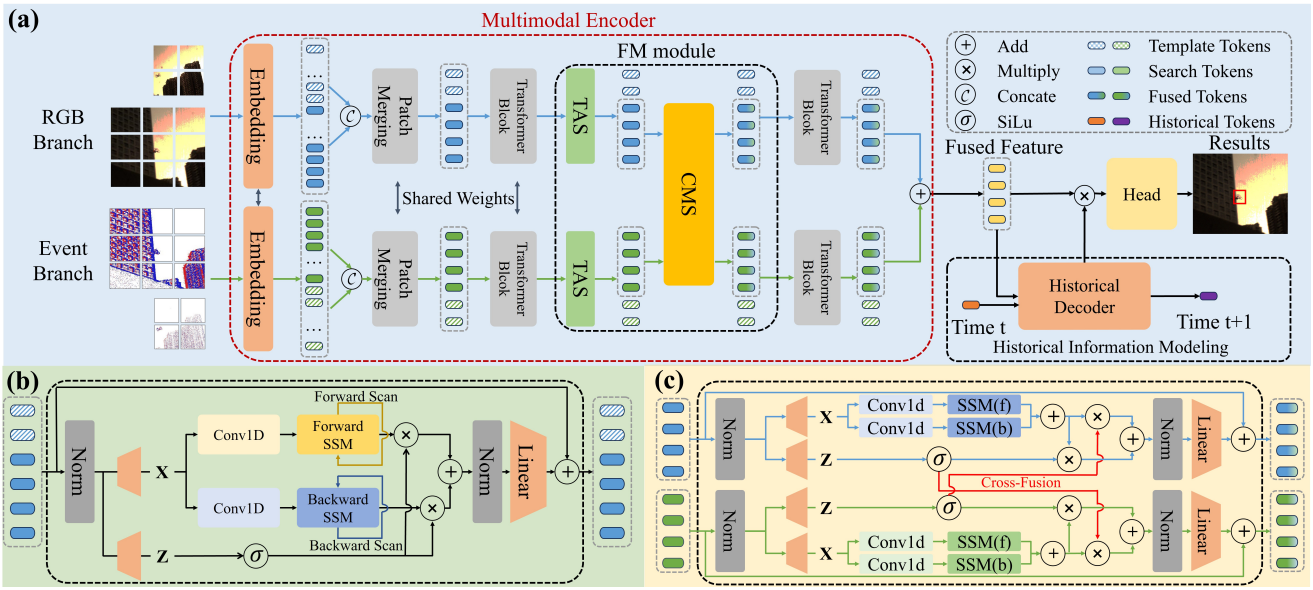


Figure 3. (a) Details of our MamTrack, which employs the Multimodal Encoder to extract and fuse the features of RGB and event modals and introduces the Historical Decoder to capture the temporal cues related to target state changes; and two key components in the Fusiom Mamba (FM) module: (b) Target-Aware Scan (TAS) and (c) Cross-Modality Scan (CMS) module.

recent studies [2, 13, 48] incorporate prompt tuning strategies into the Transformer architecture to transfer pretrained RGB-based trackers to multimodal data.

The methods mentioned above primarily emphasise multimodal fusion and overlook the significance of mining historical information, which is crucial for multimodal tracking. In contrast, we explore the great potential of Mamba for multimodal tracking tasks. By combining the gating signals of two modalities in Mamba and leveraging the temporal modeling mechanism of Mamba, our approach is capable of concurrently achieving adaptive RGB-E multimodal fusion and efficient utilization of historical cues.

2.3. Tracking with Historical Temporal Information

The historical information can provide valuable insights into the target's movements, appearance, and behaviour, thereby enhancing the tracker's confidence. Based on the number of search images input into the tracker, existing tracking approaches that leverage historical temporal cues can be categorised into two groups. The first one involves only inputting one search image and implementing a dynamic kernel filter [1, 6, 44] or a specialized update branch [4, 27, 38] to capture the target appearance changes with time and dynamically update template images during tracking. However, these methods generally involve handcrafted strategies, requiring tedious tuning for various application conditions. Another line of work [25, 35, 36, 47] takes multiple search sequences as input and then models the sequential evolution of target trajectories across frames by explicitly decoding the entire trajectory sequence step by step. Despite these advances, Transformer architectures employed in these methods are not intrinsically designed for temporal sequential modeling, which presents an inherent limitation. Competing with Transformer, Mamba [11] has shown great potential for long sequence modeling. Therefore, our research endeavors to design a novel Mamba-based historical decoder to effectively transmit temporal information in a parallel training manner.

3. Methodology

The overall architecture of our proposed MamTrack is illustrated in Figure 3, which has two branches with shared weights: RGB branch and Event branch. Each branch receives a template and multiple search images as inputs and corresponding features can be extracted by the Multimodal Encoder. The core of the Multimodal Encoder is our proposed Fusion Mamba (FM) module, which is inserted between Transformer encoder layers to leverage the advantages of both RGB and event modalities. In particular, the RGB modal offers abundant rich texture information, while the event modal is adept at handling challenging scenarios. The fused features are then input into our other primary component, the Historical Encoder, which specifically aims to capture the target appearance changes and motion trends by utilizing temporal cues. Finally, the fused features are multiplied with the decoder's output and then fed into the prediction head to locate the targets.

3.1. Multimodal Encoder

Following [36], we employ a gradual downsampling network as multimodal encoder, which uses a 4×4 patch size for patch embedding. Subsequently, we adopt a total of eight multi-layer perceptron (MLP) layers and two merging layers for gradual downsampling. After this, we obtain the RGB template tokens $F_z^R \in \mathbb{R}^{N_z \times D}$, RGB search tokens $F_x^R \in \mathbb{R}^{N_x \times D}$, event template tokens $F_z^E \in \mathbb{R}^{N_z \times D}$, and event search tokens $F_x^E \in \mathbb{R}^{N_x \times D}$. Here, $N_z = \frac{H_z W_z}{16^2}$; $N_x = \frac{H_x W_x}{16^2}$; $D = 512$; $H_z W_z$ and $H_x W_x$ represent the size of the template and search images, respectively. Then, the template tokens and search tokens are concatenated and fed into the N-layer Transformer-based encoder for joint modeling. The operation of our encoder is defined as:

$$F_0^m = \mathcal{C}(F_z^m, F_x^m), \quad m \in \{RGB, Event\} \tag{3}$$

$$F_n^R, F_n^E = \mathcal{E}_n(F_{n-1}^R, F_{n-1}^E), \quad n = 1, \ldots, N \tag{4}$$

where $\mathcal{C}$ is the element-wise concatenation; $\mathcal{E}_n$ denotes n_{th} encoder layer; F_n^R and F_n^E are the RGB and event outputs of the n_{th} layer, respectively. The above operations facilitate the interaction between the template and search tokens during the RGB and event feature extraction. Next, we describe in detail our Mamba-based fusion module inserted between encoding layers.

3.2. Fusion Mamba Module

Our proposed Fusion Mamba (FM) module aims to fuse multimodal semantic information. The key to FM is designed based on the following observations: (i) The interaction between template and search features is crucial as it can help the tracker to learn and prioritize the target-relevant search features; (ii) Directly combining RGB and event modals disregards cases where one of modalities cannot provide valuable cues. In the FM, we present Target-Aware Scan module (TAS) and Cross-Modality Scan module (CMS) to address the above issues, respectively.

Target-Aware Scan Module (TAS). The TAS module enhances the interaction between template and search regions by leveraging Mamba's global modeling capability. Compared with the self-attention in Transformer, our TAS employs the bidirectional SSM mechanism [49] to estimate the relevance between all template tokens and each search token. The completeness of template cues can encourage our tracker to focus on informative search tokens. Given $F_n^m \in \{F_n^R, F_n^E\}$, as shown in Figure 3 (b), the TAS module is formulated as:

$$X_n^m = \mathcal{L}(\mathcal{N}(F_n^m)), Z_n^m = \sigma(\mathcal{L}(\mathcal{N}(F_n^m))), \tag{5}$$

$$Y_f = \mathcal{SSM}_f(\psi(X_n^m)), Y_b = \mathcal{SSM}_b(\psi(X_n^m)), \tag{6}$$

$$H_n^m = \mathcal{L}(\mathcal{N}(Z_n^m \odot Y_f + Z_n^m \odot Y_b)) + F_n^m, \tag{7}$$

where $\mathcal{L}$ is Linear projection; $\mathcal{N}$ denotes LayerNorm operation; σ is SiLU activation function; ψ is 1D convolution operation with the kernel size of 4; $\mathcal{N}$ means element-wise multiplication; X_n^m is the projection of F_n^m; Z_n^m is gate signal; Y_f and Y_b are the outputs through forward and backward directions of SSM (*i.e.*, $\mathcal{SSM}_f$ and $\mathcal{SSM}_b$). $H_n^m \in \{H_n^R, H_n^E\}$ is the enhanced output of our TAS module in the n_{th} encoder layer.

Cross-Modality Scan Module (CMS). CMS is responsible for improving the relevance and complementarity of multimodal features through a simple yet effective cross-fusion scheme. Different from TAS, we only input search tokens into CMS to prevent interference from template features. For simplicity, we will use the RGB domain as an example to describe the specific operation since RGB and event domains function similarly. Given RGB search tokens $H_n^{R,x}$ in the n_{th} layer, as shown in Figure 3 (c), our CMS is defined as:

$$X_n^{R,x} = \mathcal{L}(\mathcal{N}(H_n^{R,x})), Z_n^{R,x} = \sigma(\mathcal{L}(\mathcal{N}(H_n^{R,x}))), \tag{8}$$

$$Y' = \mathcal{SSM}_f(\psi(X_n^{R,x})) + \mathcal{SSM}_b(\psi(X_n^{R,x})), \tag{9}$$

$$Y_n^{R,x} = \mathcal{L}(\mathcal{N}(Y' \odot Z_n^{R,x} + Y' \odot Z_n^{E,x})) + H_n^{R,x}. \tag{10}$$

The modality-specific gating signals $Z_n^{R,x}$ and $Z_n^{E,x}$ are utilized to direct the interaction between modalities, thereby strengthening the primary modality while suppressing the degraded modality. With the assistance of our proposed TAS and CMS, the multimodal encoder can efficiently facilitate the interaction between template and search tokens, as well as between RGB and events modalities. The RGB and event features of the last encoder layer are added together to get the final fused feature $\mathcal{F}_s$, which will be used to enhance the functionality of the prediction head and historical decoder.

3.3. Historical Decoder

Historical temporal information is crucial in visual object tracking as it provides the state changes of targets across frames. To fully exploit the temporal cues, we propose the Historical Decoder with M layers, each layer consists of two key components: Historical State Aware (HSA) module and Feature Sequence Attention (FSA) module.

Historical State Aware Module (HSA). Mamba offers a natural advantage over conventional CNNs and Transformers when it comes to long sequence modeling. We thus present Mamba-based HSA module to capture the target historical states of N frames within a sequence into N one-dimensional learnable vectors $Q(q_1, \ldots, q_N) \in \mathbb{R}^{N \times D}$. Mamba models the sequence and outputs $Q' = (q', \ldots, q'_N) \in \mathbb{R}^{N \times D}$, this is

$$X_Q = \mathcal{SSM}(\psi(\mathcal{L}(Q))), Z_Q = \sigma(\mathcal{L}(Q)), \tag{11}$$

$$Q' = \mathcal{L}(\mathcal{N}(X_Q \odot Z_Q) + Q. \tag{12}$$

The output Q' will be subsequently utilized as the input of FSA. It is important to highlight that, with the help of the

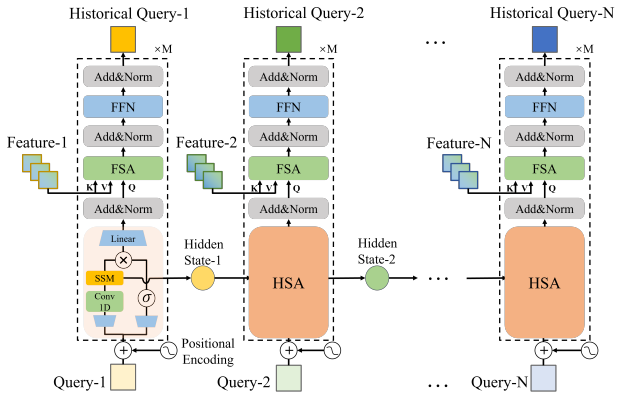


Figure 4. Overview of our Historical Decoder with two key components: Historical State Aware (HSA) module, and Feature-Sequence Attention (FSA) module.

SSM mechanism, the hidden state generated by each query at the current moment will subsequently be transmitted and contribute to the estimation in the following moment. In this way, we guarantee the long-term viability of modeling historical cues. Compared to Transformer-based autoregressive models, this significantly increases both inference and training speeds.

Feature-Sequence Attention Module (FSA). Our proposed FSA module aims to combine multimodal spatial features $\mathcal{F}_s$ and historical temporal features Q' to improve the confidence of the prediction head. Our FSA module employs the multi-head attention mechanism, where historical information sequences are used as queries and multimodal features are used as keys and values:

$$\begin{aligned} Q'' &= \mathcal{N}(\mathcal{C}((H_1, \ldots, H_{n_h})W^O)) + Q', \\ H_i &= \mathcal{ATTN}(Q'W_i^Q, \mathcal{F}_s W_i^K, \mathcal{F}_s W_i^V), \\ \mathcal{ATTN}(q,k,v) &= \varphi\left(\frac{qk^\top}{\sqrt{d_k}}\right)v, \end{aligned} \tag{13}$$

where φ is the Softmax function; $W_i^Q \in \mathbb{R}^{d_m \times d_k}$, $W_i^K \in \mathbb{R}^{d_m \times d_k}$, $W_i^V \in \mathbb{R}^{d_m \times d_v}$, and $W^O \in \mathbb{R}^{n_h \times d_m}$ are all learnable parameters. In our method, we employ multi-head attention with 8 heads, *i.e.*, $n_h = 8$, $d_m = 512$, and $d_k = d_v = \frac{d_m}{8} = 64$. Then Q'' is fed into the following decoder layer to generate the final temporal queries $\mathcal{Q}$ after being processed by the Feed Forward Network (FFN). Finally, we further combine the spatial and temporal information without introducing any parameters by:

$$F_{out} = \mathcal{F}_s \odot \mathcal{Q}^\top, \tag{14}$$

where F_{out} represents the multimodal features enriched with historical temporal information, and highlights the position where the target may be located. Then it will be fed into the tracking head for the target bbox prediction.

3.4. Head and Loss

We employ a center-based head as regular [39] to predict the center position and scale of the target. The prediction head outputs include the classification score map, the bounding box size, and the offset size. We use focal loss [19] and GIoU loss [24] for training the network. The total loss L is:

$$L = L_{cls} + \lambda_{iou} L_{iou} + \lambda_{L_1} L_{L_1}, \tag{15}$$

where L_{cls} denotes the weighted focal loss for classification, the generalized GIoU loss L_{iou} and L1 loss L_1 are adopted for bounding box regression, $\lambda_{iou} = 2$ and $\lambda_{L_1} = 5$ are the hyperparameters.

4. Experiments

4.1. Experimental Settings

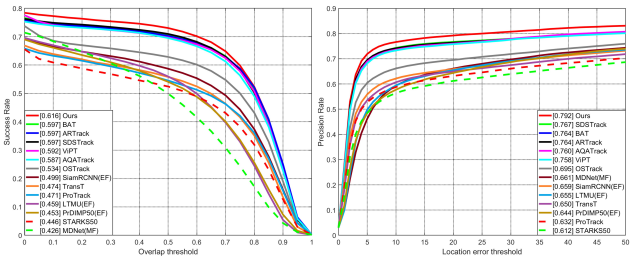
Implementation Details. We implemented MamTrack using PyTorch and trained it on 8 NVIDIA RTX 4090 GPUs with a batch size of 16. We employed the AdamW optimizer with a weight decay of 10^{-4} and a learning rate of 4×10^{-4}. We utilized pretrained HiViT-Base [45] as the Transformer encoder with 20 layers. Fine-tuning MamTrack on the RGB-Event training set took 50 epochs. Our proposed Fusion Mamba module is inserted into the 12th, 15th, and 18th layers. The decoder consists of 3 layers (*i.e.*, $M = 3$). Besides, we utilize the Hamming window to incorporate positional priors.

Evaluation Metrics. We adopt three widely used metrics for quantitatively evaluating the tracking performance: Success Rate (SR), Precision Rate (PR), and Overlap Precision (OP_T). The SR focuses on the frame where the overlap between ground truth and predicted bounding box is larger than a threshold; OP_T matches SR but with a specific overlap threshold T. The PR metric counts the frame on which the center distance between the ground truth and predicted bounding box is within a given threshold.

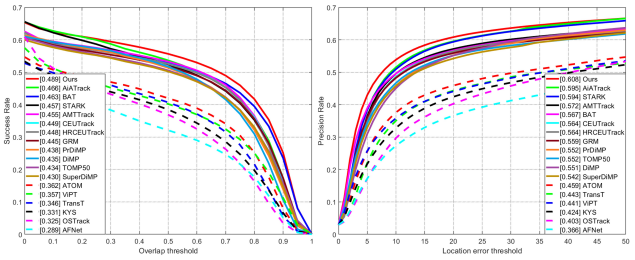
4.2. Comparisons with State-of-the-arts

To demonstrate the effectiveness of our proposed approach, we compare our tracker with representative state-of-the-art trackers on three widely utilized benchmarks: VisEvent [29], FELT [31], and FE108 [41].

VisEvent. VisEvent is a large-scale visible-event tracking benchmark, which contains 500 training and 320 testing sequences with 17 attributes. As shown in Figure 5(a), our MamTrack achieves the top SR and PR of 61.6% and 79.2%, respectively, which beats all single- and multi-modal SOTA trackers. For example, our method outperforms the pure Transformer-based temporal tracker AQATrack [36] by 2.9% on SR and multimodal tracker SDSTrack [13] by 1.9%. Figure 6 further provides the tracking performance under 17 challenging attributes. We can see



(a) Success (left) and Precision (right) plot on VisEvent



(b) Success (left) and Precision (right) plot on FELT

Figure 5. Comparison of Success (left) and Precision (right) plots on VisEvent (a) and FELT (b).

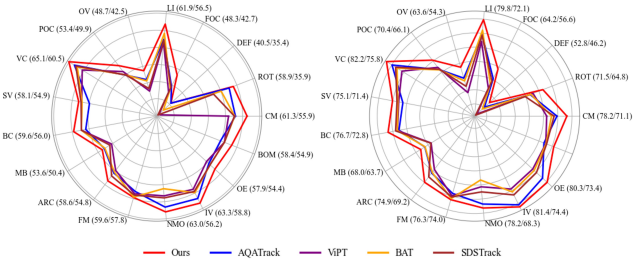


Figure 6. Comparison of SR (left) and PR (right) across various attributes using our approach against four top trackers on VisEvent, highlights the performance of Ours/Lowest.

that our approach still fares best in all degraded conditions, demonstrating its ability to effectively fuse the strengths of RGB and Events.

FELT. FELT is a large-scale long-term frame-event single object tracking dataset, which contains 742 videos, 1,594,474 RGB frames and corresponding events. The results in Figure 5(b) show that our MamTrack achieves a 48.9% overall SR and 60.8% PR. The results illustrate that our approach achieves effective long-term tracking by exploiting the historical temporal modeling capability of Mamba. We also offer an attribute-based evaluation in the supplementary material for further analysis, our MamTrack achieves the best performance on most attribute splits.

FE108. FE108 is a visible-event multimodal tracking dataset of indoor scenes, which contains 76 videos for training and 32 for testing. As shown in Table 1, our MamTrack outperforms the previous best tracker ViPT by 0.6% in SR and 0.4% in PR. It demonstrates that our approach is also highly competitive for short-term tracking scenarios.

ATOM	DiMP	PrDiMP	FENet	MixFormer	TransT
52.8/80.0	53.4/88.2	55.2/86.8	63.4/92.4	60.4/87.8	56.7/89.0
SiamFc++	STARK	OSTrack	ViPT	HDETrack	**Ours**
54.5/85.3	55.4/83.7	57.1/89.3	65.8/93.8	59.8/92.2	**66.4/94.2**

Table 1. Quantitative comparison of SR(%) / PR(%) against SOTA trackers on the FE108 dataset.

#	RGB	Event	TAS	CMS	CF	FSA	HSA	SR(%)	PR(%)
1	✗							40.5	56.1
2		✗						59.0	75.8
3			✗					60.7	78.1
4				✗				60.2	77.4
5					✗			60.0	77.1
6						✗		60.6	78.0
7							✗	60.4	77.8
8								**61.6**	**79.2**

Table 2. Ablation study for important components on VisEvent. Blank denotes the component is used by default, while ✗ represents the component is removed.

4.3. Ablation Study and Analysis

Impact of Multimodal Input. We assess the effectiveness of multimodal fusion by separately utlizing event-only and RGB-only inputs. As shown in Table 2 #1 and #2, the tracking results with unimodal input are significantly worse than those with multimodal input, demonstrating that our method effectively utilizes both modalities' advantages to improve tracking under degraded conditions.

Effectiveness of Fusion Mamba Module. There are two key mamba-based components in our Fusion Mamba (FM) module: (i) Target-Aware Scan (TAS) facilitates the interaction between the template and search tokens; (ii) Cross-Modality Scan (CMS) focuses on the fusion of event and RGB modalities in an adaptive manner. To verify their effectiveness, we modify the original model by dropping each of the components and retraining the modified models. The tracking performance in Table 2 #3 and #4 has shown a decrease to varying degrees, indicating the effectiveness and rationality of our designed FM module. In CMS, the gate-based operation Cross-Fusion (CF) is the primary operation that supports the interaction between the two modalities. We thus eliminate the CF to showcase its indispensability. #5 shows that when the CF is removed, the SR drops by 1.6%. This illustrates that the deep interaction among features from both modalities plays a key role.

One possible factor that impacts the contribution of FM to our tracker is the location of FM insertion in the encoder. To verify this, several additional sets of experiments were

Location	SR (%)	PR(%)	Location	SR (%)	PR(%)
12	60.0	77.0	14-17-20	61.1	78.4
12-15	61.0	78.3	12-13-14	60.1	77.3
12-15-18 (Ours)	**61.6**	**79.2**	18-19-20	60.6	77.9

Table 3. Comparison of different FM module locations on VisEvent. x-y-z represents our FM module is inserted into the xth, yth, and zth encoder layers.

Type of HSA	Number of Layers	SR(%)	PR(%)	FPS
None	0	59.9	76.8	20.7
Mamba	1	60.5	78.1	19.9
	3	**61.6**	**79.2**	**18.1**
	6	60.9	78.1	16.6
Transformer	3	61.0	78.4	16.9

Table 4. Ablation studies on the type and number of decoder layers.

conducted. As shown in Table 3, compared to other 3-stage insertion styles, the 12-15-18 style adopted in this paper provides the best performance, which can be attributed to our insertion strategy that focuses on both low-level texture features and high-level semantic features. Moreover, the tracking performance exhibits a progressive improvement from 12 to 12-15 and then to 12-15-18 style, revealing the growing synergy between RGB and event modalities facilitated by our FM module.

Effectiveness of Historical Decoder. The Historical Decoder is designed to capture and transmit historical information during tracking, which consists of two main components: Historical State Aware (HSA) and Feature Sequence Attention (FSA) modules. As shown in Table 2 #6, when the FSA is removed, the SR score drops by 1.0%, showing the importance of multimodal features for exploiting historical information. The performance drops by 1.2% when the HSA is removed as shown in #7. This illustrates the effectiveness of Mamba for modeling long sequences.

We also examine the influence of the number of historical decoder layers on tracking performance. As shown in Table 4, the utilization of the historical decoder greatly enhances the results, with the most performance observed when employing 3 decoder layers. To further verify the effectiveness of the Mamba in the historical decoder, we replace the Mamba architecture with Transformer. Table 4 shows our method with Mamba excels in both performance and speed due to Mamba's linear complexity and its capability to model long sequences.

Impact of the Length of Historical Information. As shown in Table 5, the model's performance improves as the range of historical information increases from 1 to 4. This

Length	SR(%)	PR(%)	Length	SR(%)	PR(%)
1	60.1	77.5	8	59.8	76.9
2	60.4	77.9	16	59.7	76.6
4	**61.6**	**79.2**			

Table 5. Impact of the length of historical information.

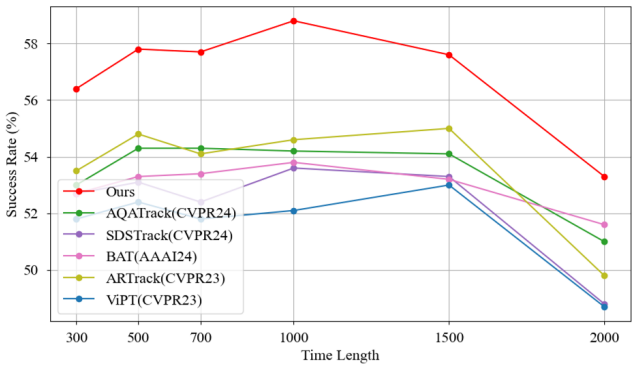


Figure 7. Comparison of SR scores for long-term tracking on VisEvent, where the horizontal axis is test sequence length.

indicates that the model can effectively capture the target's movement trends and appearance changes, resulting in improved performance. However, when the historical information range is extended to 8 and 16, the model's performance slightly declines. This decrease is due to the limited number of video sequences used in each iteration, which impairs the model's generalization ability.

Effectiveness for Long-term Tracking. To further demonstrate the effectiveness of our Mamba-based historical decoder in capturing temporal information, we analyze its performance in long-term tracking. Specifically, we evaluate the tracking performance of sequences containing more than i frames on the VisEvent individually, where $i \in \{300, 500, 700, 1000, 1500, 2000\}$. Figure 7 shows that our MamTrack outperforms other SOTA Transformer-based temporal trackers and multimodal trackers by a large margin in SR across all settings. This suggests our Mamba-based approach proves superior in capturing long-term cues compared to the Transformer-based autoregressive scheme.

4.4. Performance on RGB-T Tracking Dataset

To further validate the generalization ability of our MamTrack, we conducted additional experiments on the RGB-T tracking task. Following the common settings used in previous work [48], we trained our tracker on the LasHeR [16] training set and tested it on both the LasHeR test set and the RGBT234 [15] dataset.

Results on LasHeR Dataset. LasHeR is the largest RGB-T tracking dataset at present, containing 1224 aligned video sequences with diverse attribute annotations. Of

LasHeR	mfDimp	SGT++	OSTrack	ProTrack	HMFT	APFNet	CEUTrack	ViPT	**Ours**
	34.3/ 44.7	25.1/36.5	41.2/51.5	42.0/53.8	31.3/ 43.6	36.2/50.0	49.6/63.5	52.5/65.1	**54.2/67.4**
RGBT234	mfDimp	CMPP	OSTrack	ProTrack	HMFT	APFNet	MFGNet	ViPT	**Ours**
	42.8/ 64.6	57.7/82.3	54.9/72.9	57.9/82.7	56.8/78.8	57.9/82.7	53.5/78.3	61.7/ 83.5	**62.4/84.4**

Table 6. Performance comparison of SR(%) / PR(%) on LasHeR and RGBT234 dataset.

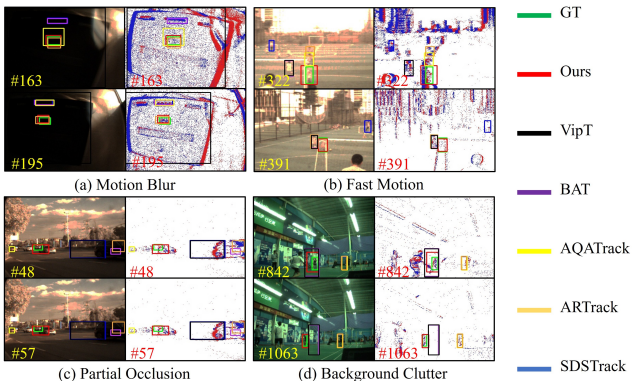


Figure 8. Visual comparison of our MamTrack against other SOTA trackers under 4 challenging conditions on VisEvent.

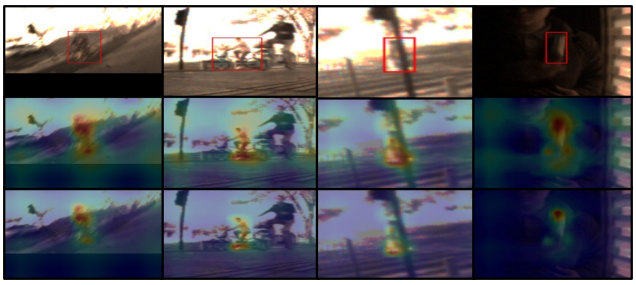


Figure 9. The comparison attention map in the MamTrack. The first row is the ground truth; the second and third rows display the attention maps in the multimodal encoder and historical decoder, respectively.

these, 245 sequences are designated as the testing set, while the remaining are used for training. As shown in Table 6, MamTrack performs exceptionally well on RGB-T tracking tasks, effectively leveraging the advantages of both thermal and visible information. This indicates its potential for other multimodal tracking tasks.

Results on RGBT234 Dataset. RGBT234 is a large-scale RGB-T tracking dataset comprising 234 pairs of videos. As shown in Table 6, our MamTrack surpasses the most advantaged tracker ViPT [48] by 0.7% in Success Rate (SR) and 0.9% in Precision Rate (PR).

4.5. Visualization

We first provide a visual comparison across four challenging conditions on the VisEvent dataset. As shown in Figure 8, our approach can accurately track the target under all conditions. In particular, our method effectively fuses the advantages of RGB and event to enhance tracker confidence when the target is in scenes with motion blur or fast motion. Besides, in situations where the target is partially occluded or the background is complex, the historical cues encourage our method to recall the target's appearance, thereby minimizing environmental interference.

Furthermore, we also visualize the attention maps of the encoder and decoder to demonstrate the effectiveness of historical information. As shown in Figure 9, guided by historical information, our MamTrack pays more attention to the target's location under challenging conditions.

5. Conclusion

We propose MamTrack, a novel approach jointly leverages multimodal spatial and historical temporal information for RGBE object tracking. Our historical decoder, based on the innovative Mamba architecture, effectively captures target appearance changes and motion trends, enabling the model to locate the target within complex backgrounds. We also introduce a simple yet effective Mamba-based fusion method for integrating cross-modality cues. Extensive evaluations demonstrate that our method achieves competitive performance compared to previous SOTA trackers. However, we primarily focuses on the effectiveness of Mamba in modality fusion, but not leverage the high temporal resolution characteristics of event data. Future research will explore this aspect to enhance the performance of the tracker.

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